

Infinitesimal Calculus 2, list of theorem for Quiz 2

1. Prove that the Dirichlet function is not integrable on any closed interval $[a, b] \subseteq \mathbb{R}$.
2. Let $a, b \in \mathbb{R}$ such that $a < b$ and let f be a function that is bounded on $[a, b]$. Suppose f is integrable on $[a, b]$ Prove

$$\left| \int_a^b f(x) dx \right| \stackrel{\Delta}{\leq} \int_a^b |f(x)| dx.$$

3. Let $a, b \in \mathbb{R}$ such that $a < b$ and let f be a function that is continuous on $[a, b]$. Prove $\exists c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

4. Let $a, b \in \mathbb{R}$ such that $a < b$ and let f be a function that is bounded on $[a, b]$. Suppose f is integrable on $[a, b]$ and consider the function defined by

$$F(x) = \int_a^x f(t) dt, \quad \forall x \in [a, b].$$

Prove that F is continuous.

5. Let $a, b \in \mathbb{R}$ such that $a < b$ and let f be a function that is bounded on $[a, b]$. Suppose f is integrable on $[a, b]$ and consider the function defined by

$$F(x) = \int_a^x f(t) dt, \quad \forall x \in [a, b].$$

Let $x_0 \in [a, b]$. Suppose f is continuous at x_0 . Prove F is differentiable at x_0 , and $F'(x_0) = f(x_0)$.

6. Let $a, b \in \mathbb{R}$ such that $a < b$ and let f, G be two functions defined on $[a, b]$. Suppose f is continuous on $[a, b]$, and that G is an anti-derivative of f on $[a, b]$. Prove that

$$\int_a^b f(x) dx = G(b) - G(a).$$

7. Let $a \in \mathbb{R}$ and let f, g be two functions defined on $[a, \infty)$. Suppose that

(i)

$$0 \leq f(x) \leq g(x), \quad \forall x \geq a.$$

(ii) f, g are integrable on $[a, b]$ for every $b > a$.

(iii) The improper integral $\int_a^\infty g(x) dx$ is convergent.

Prove that the improper integral $\int_a^\infty f(x) dx$ is convergent and

$$\int_a^\infty f(x) dx \leq \int_a^\infty g(x) dx.$$